



## PROFESSIONAL SUMMARY

Researcher and educator at the intersection of AI, autonomous systems, and aerospace infrastructure, with an interdisciplinary foundation in electrical, civil, and industrial engineering. Specializes in AI-driven digital twins and generative modeling for UAV and Advanced Air Mobility systems, including trajectory synthesis, synthetic environment generation, and air-to-ground communication modeling for next-generation airspace and defence applications. Experience spans UAV/AAM digital twins, generative AI for autonomous systems, human-centered AI design, and large-scale mobility modeling, now applied to airspace, communication, and autonomy challenges in space and defence.

## EDUCATION

**Ph.D., Civil Engineering** (Smart City Focus), 2022; University of Central Florida, Orlando, FL

Advisor: Dr. M. Zaki, Assistant Professor

Dissertation: Generative Modeling of Human Behavior: Social Interaction and Networked Coordination in Shared Facilities <https://stars.library.ucf.edu/etd2020/1381/>

**M.S., Electrical & Computer Engineering**, 2016; University of Nevada Las Vegas, Las Vegas, NV

Advisor: Dr. Pushkin Kachroo, Lincy Professor

Thesis: Complex Traffic Network Modeling & Area-Wide Hierarchical Control <https://digitalscholarship.unlv.edu/thesesdissertations/2781/>

**B. Tech., Electronics and Instrumentation Engineering**, 2009; U.P. Technical University, Lucknow, India

Advisor: Dr. Ragini Tripathi, Professor

Project: Portable Ozone Generator: Equipment Design and Analysis

## WORK EXPERIENCE

- Feb'25-present **Post-Doctoral Research Associate, Industrial Engineering and Management Systems**, University of Central Florida, Orlando
- Designing, implementing, and validating simulation and digital twin environments to enhance Air Traffic Management (ATM) capabilities for Advanced Air Mobility (AAM) systems.
  - Conducting human-machine interaction research to understand human behavior, trajectory evolution, and interaction dynamics in shared spaces.
  - Contributing to competitive proposal development (e.g., NASA MPLAN, USDOT, USRC), along with journal and conference publications.
  - Mentoring graduate and undergraduate students on multiple NASA-related projects and proposals, including innovative approaches to drone operations in GPS-denied environments.
- Mar'23-Mar'25 **Data Analyst**, Insight Transportation Consulting, Orlando
- Conducted ridership and travel time/demand forecasting using four-step travel demand models for projects including Inglewood Transit Connector, Sunshine Corridor, and CFRPM.
  - Evaluated alternative routes and conducted corridor studies, performance assessments, and multi-modal (auto, bus, rail) integration analyses for projects including BCT Transit Development Plan, Metro Purple Line BRT, LA Metro, and Metra Rail.
  - Analyzed on-board survey and APC data to inform schedule adjustments; built a Python-based GTFS Real-Time processing tool for trip cancellations and arrival predictions. Key projects: Campbellton Corridor, COTA, GTFS Enhancer.
- May'22-Nov'22 **Sustainability Intern**, Orange County Public Works, Orlando
- Worked on the Orange County Sustainable Operations and Resilience Action Plan. Prepared a *training presentation* and drafted a *white paper* for Public Works management and staff based on best practices from FDOT, FHWA, and TRB.
- 2018-2022 **Graduate Research/Teaching Assistant**, University of Central Florida, Orlando
- Conducted SAFER-SIM (Tier-1 UTC)-funded research on active mobility and urban safety.
  - Developed a trajectory planning framework to analyze interactions of active mobility users in shared urban environments.
  - The receding horizon optimization-based trajectory planning framework can simulate realistic scenarios, aiding in proactive urban planning and designing safer cities.



- Aug'17-Dec'17 **Adjunct Faculty**, Dept of Electrical & Computer Engineering, CSULA  
 • EE3001: Numerical Analysis and Modeling using MATLAB.
- Nov'16-Nov'17 **Web and Marketing Analyst**, PolyPak America, Los Angeles  
 • Data Analytics, Marketing Analytics, IT and Website Management
- Jan'14-Aug'16 **Research Assistant**, Transportation Research Center, UNLV  
 • Crash data analysis, occupant protection, and machine learning

#### FUNDED RESEARCH GRANTS AS PI or Co-PI

1. **Title:** A Digital Twin for AAM Communication Channels  
**Agency:** NASA  
**Role:** PI (90% share)  
**Amount:** \$50,000  
**Status:** **Awarded:** <https://www.ucf.edu/news/nasa-funded-ucf-research-advances-urban-air-mobility-by-mapping-radio-signals/>

Note: Additional proposals in submission across NASA, USDOT, and industry partners. Curriculum Vitae will be updated as decisions come in.

#### JOURNAL PUBLICATIONS

Google Scholar: <https://scholar.google.com/citations?user=nePcSDMAAAJ&hl=en>

1. Shaurya Agarwal, **Saumya Gupta**, Pushkin Kachroo and Nilesh Dhingra. "A machine learning based approach for smart and automated data collection: applications in transportation" *Transportation in developing economies*, 10(1), 15., (2024)
2. **Saumya Gupta**, Mohamed H. Zaki and Adan Vela. "HISS: A Human Trajectory Planning Simulation Framework for Shared Space" *IEEE Open Journal of Intelligent Transportation Systems*, (2023)
3. **Saumya Gupta**, Mohamed H. Zaki, and Adan Vela. "Generative Modeling of Pedestrian Behavior: A Receding Horizon Optimization-Based Trajectory Planning Approach." *IEEE Access* 10 (2022): 81624-81641.
4. Pushkin Kachroo, **Saumya Gupta**, Shaurya Agarwal, and Kaan Ozbay. "Optimal Control for Congestion Pricing: Theory, Simulation and Evaluation." *IEEE Transactions on Intelligent Transportation Systems*, 18, no. 5 (2017): 1234-1240.

#### BOOK CHAPTERS

1. Archie J Huang, **Saumya Gupta**, Shaurya Agarwal, "Physics-informed machine learning: Applications in smart transportation", *Machine Learning in Civil Engineering and Infrastructure Development: A Practitioner's Handbook*, M.Z. Naser, (2026)

#### FULL PAPER REVIEWED CONFERENCE PROCEEDINGS

1. **Saumya Gupta** and Adan Vela. "Diffusion-Based Synthesis of sUAS Crosstrack Deviation Trajectories for Simulation" *2026 AIAA DATC/IEEE 45<sup>th</sup> Digital Avionics Systems Conference (DASC), Orlando, USA, 2026. (Accepted)*
2. **Saumya Gupta** and Adan Vela. "Generation of Synthetic Cities for AAM Research" *2026 AIAA DATC/IEEE 45<sup>th</sup> Digital Avionics Systems Conference (DASC), Orlando, USA, 2026. (Accepted)*
3. **Saumya Gupta** and Adan Vela. "A Modular Ray-Tracing Framework for Simulating Air-to-Ground Communication in Urban Environments." *2025 AIAA DATC/IEEE 44<sup>th</sup> Digital Avionics Systems Conference (DASC), Montreal, Canada, 2025.*
4. Syed Mostaqim Ali, **Saumya Gupta** and Mohamed H. Zaki. "Deep Reinforcement Learning-Based Pedestrian Trajectory Generation: Incorporating Social Context." *103<sup>rd</sup> Transportation Research Board (TRB) Annual Meeting, Washington DC, USA, 2024.*



5. Saumya Gupta, Mohamed H. Zaki and Adan Vela. "A Receding Horizon Optimization to generate Pedestrian Trajectories in Shared Space. " 102<sup>nd</sup> *Transportation Research Board (TRB) Annual Meeting, Washington DC, USA, 2023.*
6. **Saumya Gupta**, Pushkin Kachroo, Shaurya Agarwal, and Kaan Ozbay "Exploring PageRank Algorithm and Voronoi Diagrams for Dynamic Network Partitions Facilitating Feedback Linearization based Control" 14<sup>th</sup> *International Conference on Transportation Planning and Implementation Methodologies for Developing Countries (TPMDC), IIT Bombay, 2022*

## CONFERENCE PRESENTATIONS

1. Saumya Gupta, Dr. M.Zaki. "A Simulation Platform for Evaluating Ad-Hoc Networks in Shared Facilities using Video-based movement trajectories," *ASCE International Conference on Transportation & Development (ICTD 2020)*, Seattle, May 26-29, 2020.
2. Saumya Gupta, Dr. M.Zaki. "Developing a Pedestrian Behavioral Mobility Model in Urban Facilities," *International Conference on Smart Tourism, Smart Cities and Enabling Technologies (The Smart Conference)*, Orlando, 1-4 May, 2019.
3. Saumya Gupta, Dr. M.Zaki, Dr. Hassan. "Evolutionary Dynamics of Social Media Activity during Natural Disasters," *Southern Data Science Conference and Data Science Summit: Smart Cities of the Future*, Orlando, 19 October 2018.

## SKILL SETS

|                           |                                                                                                                                                                                                                        |
|---------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Programming and Libraries | Python, MATLAB, R, Scikit-learn, PyTorch, Matplotlib, Pandas, Seaborn                                                                                                                                                  |
| Technical Strength        | Machine Learning (LSTM, NLP, Reinforcement Learning, Regression), Generative AI modeling, Applied Statistics, Data Analysis, Analytical Modeling, Mixed Integer Programming, Receding Horizon Control, Optimal Control |
| Tools                     | Simulation (VISSIM, SUMO, OMNeT++, STOPS), Visualization (ArcGIS, GTFSed, Cube), Analytical (Excel, Google Analytics, Tableau), Text Processing (Word, PowerPoint, LaTeX, yED)                                         |
| Extracurricular interests | Yoga, Cooking, Gardening, Painting, Reading non-fiction books, thoughtful conversations, and learning new skill.                                                                                                       |

## PROFESSIONAL ACTIVITIES

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|-------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Presented in            | <ul style="list-style-type: none"> <li>• Safer-Sim (UTC) Webinar 2021, <i>Improving mobility for non-motorized and vulnerable Road-Users</i></li> <li>• International Conference on Smart Tourism, Smart Cities and Enabling Technologies, Orlando, 2019.</li> <li>• Southern Data Science Conference &amp; Summit: Smart Cities of the Future, Orlando, 2018.</li> </ul> |
| Reviewer for            | <ul style="list-style-type: none"> <li>• Transportation Research Board Annual Conference</li> <li>• IEEE Intelligent Transportation Systems Conference</li> <li>• IEEE Systems, Man, and Cybernetics Conference</li> </ul>                                                                                                                                                |
| Data Mining Competition | • CFE 2018 Lending Analytics Competition                                                                                                                                                                                                                                                                                                                                  |
| Awards                  | <ul style="list-style-type: none"> <li>• UCF Doctoral Research Support Award 2021, UCF College of Graduate Studies Presentation Fellowship, UNLV Access Grant-Grad NN (multiple times)</li> </ul>                                                                                                                                                                         |

## PERSONAL DETAILS

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| VISA Sponsorship | • None required; U.S. Permanent Resident |
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## PROFESSIONAL REFERENCES

**Dr. Adan E. Vela**  
Assistant Professor, UCF Industrial Engineering & Management Systems  
University of Central Florida,



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Western University

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***Dr. Pushkin Kachroo***

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